E-classroom	**Name: Memorandum**					
	Gr 10		**Date:**		**Time:**	**1 HR**
CAPS Reference	**Functions**					
Topic	**Linear Function**					

2.1.1 $f(x) = 4x + 2$

$\underline{y - intercept; x = 0}$

$y = 4(0) + 2$

$= 2 \qquad (0; 2)$

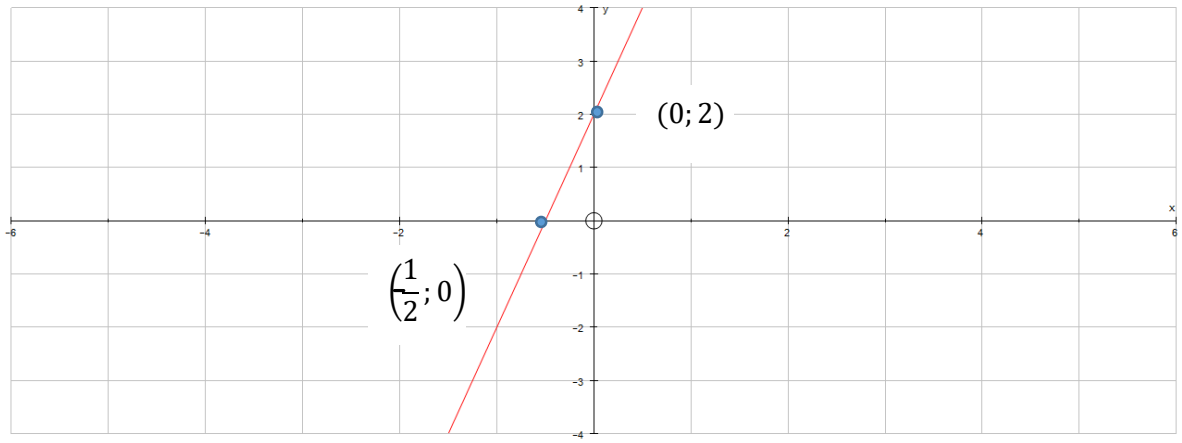
$\underline{x - intercept; y = 0}$

$0 = 4x + 2$

$-2 = 4x$

$\frac{-2}{4} = x$

$-\frac{1}{2} = x \qquad \left(-\frac{1}{2}; 0\right)$



a) $x \in (-\infty; \infty)$
b) 4
c) $x < -\frac{1}{2}; x \in R$

2.1.2 $g(x) = -2x + 1$

$\underline{y - intercept; x = 0}$

$y = -2(0) + 1$

$= 1 \qquad (0; 1)$

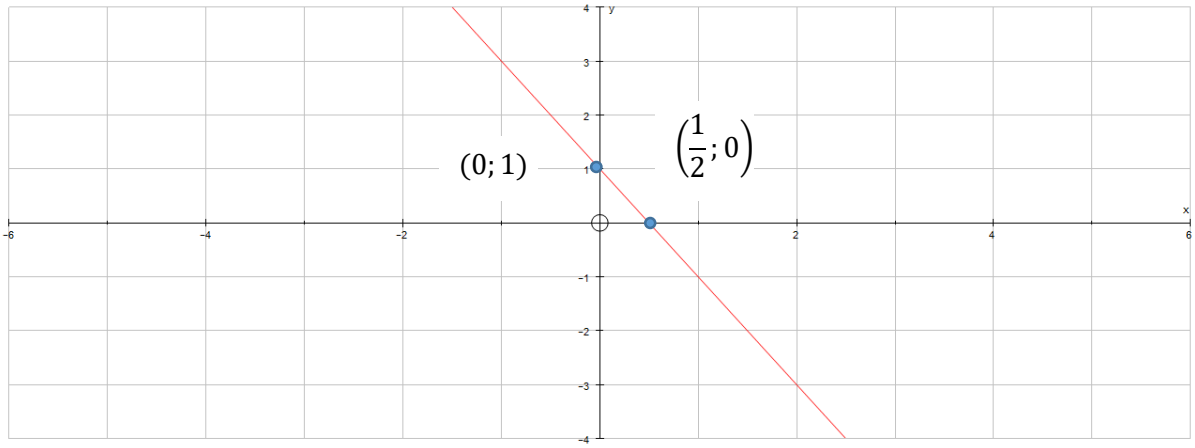
$\underline{x - intercept; y = 0}$

$0 = -2x + 1$

$-1 = -2x$

$\frac{-1}{-2} = x$

$\frac{1}{2} = x \qquad \left(\frac{1}{2}; 0\right)$



a) $y \in (-\infty; \infty)$
b) -2
c) $x < \frac{1}{2}; x \in R$

2.1.3 $h(x) = -x - 1$

$\underline{y - intercept; x = 0}$

$y = -(0) - 1$

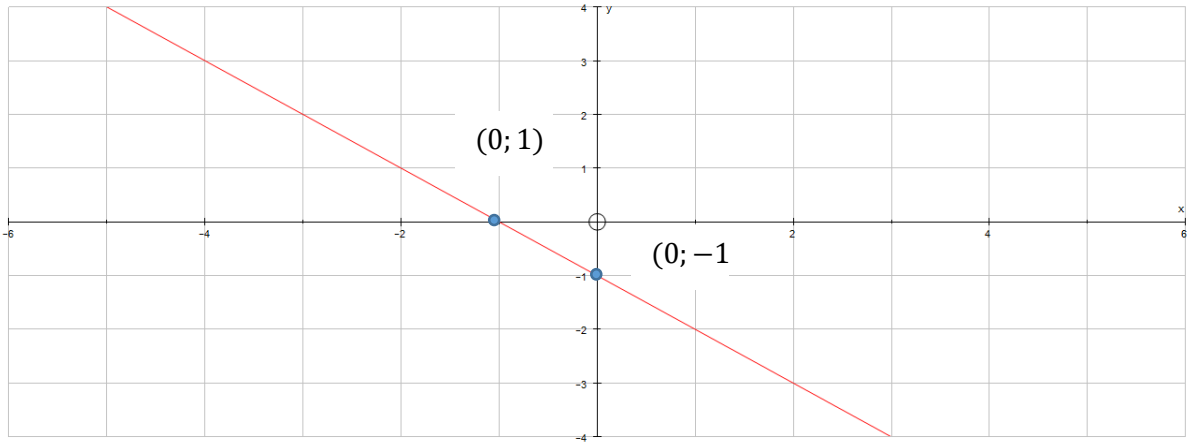
$= -1 \qquad (0; -1)$

$\underline{x - intercept; y = 0}$

$0 = -x - 1$

$1 = -x$

$-1 = x \qquad (-1; 0)$



a) $x \in (-\infty; \infty)$
b) -1
c) $x < -1 ; x \in R$

2.1.4 $k(x) = \frac{1}{3}x + 4$

$\underline{y - intercept; x = 0}$

$y = \frac{1}{3}(0) + 4$

$= 4 \quad (0; 4)$

$\underline{x - intercept; y = 0}$

$0 = \frac{1}{3}x + 4$

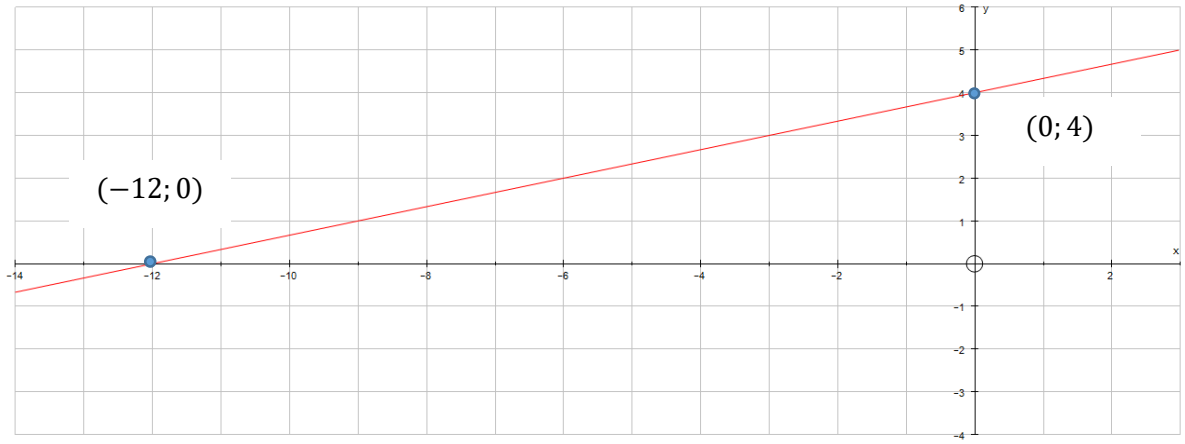
$-4 = \frac{1}{3}x$

$-4 \div \frac{1}{3} = x$

$-4 \times \frac{3}{1} = x$

$-12 = x$

$(12; 0)$



a) $y \in (-\infty; \infty)$

b) $\frac{1}{3}$

c) $x < -12 ; x \in R$

2.2.1 Given: $(0;2)\ and\ (1;-1)$

Step 1: Calculate the gradient

$x_1\ \ y_1$ $\qquad\qquad$ $x_2 y_2$

$(\ 0\ ;\ 2)$ $\qquad\qquad$ $(1;\ -1)$

$$Gradient = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-1-2}{1-0}$$

$$= \frac{-3}{1} = -3$$

Step 2: Substitute $m = -3$

$$f(x) = -3x + c$$

Step3: Substitute $(0;2)\ in\ f(x) = -3x + c$

$2 = -3(0) + c$

$2 = 0 + c$

$c = 2$

$\therefore f(x) - 3x + 2$

2.2.2 Given: $(2;0)\ and\ (0;-1)$

Step 1: Substitute $y - intercept\ \ (0,-1)$

$y = mx + c$

$y = mx - 1$

Step 2: Substitute $(2;0)\ in\ y = mx - 1$

$0 = m(2) - 1$

$1 = 2m$

$\frac{1}{2} = m$

$\therefore f(x) = \frac{1}{2}x - 1$

2.2.3 Given: $(0;5)\ and\ (-2;3)$

Step 1: Calculate the gradient

$x_1\ \ y_1$ $\qquad$ $x_2 y_2$

$(\ 0\ ;\ 5)$ $\qquad$ $(-2;3)$

$$Gradient = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{3-5}{-2-0}$$

$$= \frac{-2}{-2} = 1$$

Step 2: Substitute $m = 1$

$$f(x) = 1x + c$$

Step3: Substitute $(0;5)\ in\ f(x) = 1x + c$

$5 = 1(0) + c$

$5 = 0 + c$

$c = 5$

$\therefore f(x) = x + 5$

2.2.4 Given: $(-2;2)\ and\ (0;-3)$

Step 1: Substitute $y - intercept\ \ (0,-3)$

$y = mx + c$

$y = mx - 3$

Step 2: Substitute $(-2;2)\ in\ y = mx - 3$

$2 = m(-2) - 3$

$-2m = 2 + 3$

$-2m = 5$

$m = -\frac{5}{2} = -2\frac{1}{2}$

$\therefore f(x) = -2\frac{1}{2}x - 3$